

Nucleus AI — Phase 2 Completion Report

Phase: 2 — Context Vault, AI Content Generation, Campaign Workflows & Analytics

Status:  Complete

Date: June 2026

1. Overview

Phase 2 builds the core content-production loop of Nucleus AI on top of the Phase 1 foundation (auth, RBAC, brands, admin). Users can now upload brand knowledge into a **Context Vault**, generate **on-brand AI content**, organize that content into **campaigns**, and review **usage analytics** — all governed by the existing role-based permission system.

This report covers the **final Phase 2 increment**: Campaign Workflows, Usage Analytics, and end-to-end integration verification.

2. Features Completed (Final Increment)

A. Campaign Workflows

- **Campaign detail page** (`/campaigns/[id]`):
 - Campaign details (name, description, brand, industry, creator, status badge).
 - **Generate Content** button that opens the AI generator pre-filled with the campaign's brand and campaign context (`?campaignId&brandId&campaignName`).
 - List of all generated content for the campaign, with type, token/cost metadata, expandable content view, and copy-to-clipboard.
 - Per-campaign metric strip: generations, tokens used, estimated cost.
 - **Status workflow controls**: DRAFT → ACTIVE → PAUSED → COMPLETED → ARCHIVED (gated behind `campaign:update`).
- **Campaigns list page** (`/campaigns`) reworked to be first-party (Prisma) backed: create campaigns under a brand and navigate into each campaign's detail page.
- **AI generator** updated to honor campaign context — content generated from a campaign is automatically linked to that campaign.

B. Usage Analytics

- **Analytics page** (`/analytics`) extended with a first-party **Usage Analytics** section (the existing attribution charts are preserved below it):
 - Metric cards: **Total Campaigns**, **Total Generations**, **Total Tokens Used**, **Total Cost**.
 - **Recent Generations** table (type, brand, tokens, date).
 - **Top Brands by Usage** table (generations, tokens, cost).

C. New API Endpoints (all RBAC-guarded, Zod-validated, audit-logged)

Endpoint	Method	Permission	Purpose
/api/campaigns	GET	campaign:read	List campaigns (scoped by ownership)
/api/campaigns	POST	campaign:create	Create a campaign under a brand
/api/campaigns/[id]	GET	campaign:read	Campaign detail + aggregates
/api/campaigns/[id]/content	GET	campaign:read	Generations linked to a campaign
/api/campaigns/[id]/status	PATCH	campaign:update	Change campaign status
/api/analytics	GET	authenticated	Usage aggregations (scoped)

D. Integration Testing

Validated the full flow **user** → **brand** → **campaign** → **content generation** and verified data relationships, permissions, and aggregations.

3. What Works (Verified)

- ✓ `npx tsc --noEmit` — no type errors.
- ✓ `npm run build` — production build succeeds; all new routes compiled.
- ✓ **AuthZ**: every new endpoint returns **401** when unauthenticated (`/api/campaigns`, `/api/analytics`, `/api/campaigns/[id]`, `/content`, `/status`).
- ✓ **RBAC**: writes require `campaign:create` / `campaign:update`; non-admins are scoped to campaigns/generations belonging to brands they own; ADMIN+ see all.
- ✓ **DB relationships** (verified via a Prisma integration script, data cleaned up):
 - Brand → Campaign → AIGeneration links resolve correctly.
 - `campaign._count.aiGenerations` and token/cost `aggregate` are accurate.
 - Status transition `DRAFT` → `ACTIVE` persists.
 - `groupBy(brandId)` powers the Top Brands table correctly.
- ✓ Campaign content auto-links when generated from a campaign's "Generate Content".

4. Architecture Notes

- **Two campaign systems exist** in the codebase:
 - The **Prisma Campaign model** (frontend, first-party) — used by AI generations, the new campaign detail/list pages, and analytics. **This is the canonical system for Phase 2 content workflows.**
 - A separate **FastAPI backend** Campaign /attribution system (spend, revenue, ROAS) — still surfaced in the **Attribution Analytics** section of /analytics . The campaigns list page was migrated from the backend API to the Prisma API so that campaigns, generations, and analytics share one consistent data source.
 - All new routes follow the established Phase 1 conventions: `requirePermission` / `requireAuth` guards, `Zod safeParse` + `firstZodError` , `sanitizeText` on free text, best-effort `recordAudit` , and `export const dynamic = 'force-dynamic'` .
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5. Known Limitations (Intentionally Out of Scope)

- No advanced charts/visualizations for usage analytics (tables + metric cards only).
 - No CSV export of analytics or content.
 - No campaign duplication/cloning or bulk actions.
 - No detailed per-channel/per-period breakdowns for usage analytics.
 - Campaign editing (name/description) beyond status is not exposed in the UI yet.
 - The Prisma campaign system and the FastAPI attribution campaign system remain independent (not reconciled into a single model).
 - AI generation uses `gpt-4o-mini` ; without an `OPENAI_API_KEY` it falls back to a mock generator (clearly flagged in the UI).
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6. Phase 3 Readiness

The platform is ready for Phase 3 work. Suggested next steps:

- Reconcile / unify the Prisma campaign model with the backend attribution campaigns.
 - Add campaign editing, scheduling, and richer content lifecycle (publish/export).
 - Expand analytics with time-series charts, filters, and CSV export.
 - Wire real attribution data (spend/conversions) to generated content for ROI.
 - Add team-level dashboards and notifications around campaign status changes.
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7. Test Commands

```
cd frontend
npx tsc --noEmit          # type check
npm run build              # production build
npm run start              # serve on :3000 (VM)
# Unauthenticated endpoints return 401:
curl -i http://localhost:3000/api/campaigns
curl -i http://localhost:3000/api/analytics
```